# PRACTICAL 1-A

## Object Oriented Programming

## AIM:

To implement and understand the concepts specified in Practical 1 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Vending Machine, Toll Booth, RPSLS and Part B: MiniBank menu shell.

## CODE SCREENSHOT:

RPSLS.java 1

Part A > Practical 1-A > RPSLS.java > ...

```java
import java.util.*;

public class RPSLS {
    enum Move {
        ROCK, PAPER, SCISSORS, LIZARD, SPOCK
    }

    static int winner(Move a, Move b) {
        if (a == b)
            return 0;
        boolean beats = switch (a) {
            case ROCK -> b == Move.SCISSORS || b == Move.LIZARD;
            case PAPER -> b == Move.ROCK || b == Move.SPOCK;
            case SCISSORS -> b == Move.PAPER || b == Move.LIZARD;
            case LIZARD -> b == Move.SPOCK || b == Move.PAPER;
            case SPOCK -> b == Move.SCISSORS || b == Move.ROCK;
        };
        return beats ? 1 : -1;
    }

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Random random = new Random();
        int player = 0, computer = 0;
        for (int i = 1; i <= 5; i++) {
            System.out.print("Round " + i + " (ROCK/PAPER/SCISSORS/LIZARD/SPOCK): ");
            Move p = Move.valueOf(sc.next().toUpperCase());
            Move c = Move.values()[random.nextInt(Move.values().length)];
            int result = winner(p, c);
            if (result > 0)
                player++;
            else if (result < 0)
                computer++;
            System.out.println("You: " + p + ", Computer: " + c + " -> "
                    + (result == 0 ? "Tie" : result > 0 ? "You win" : "Computer wins"));
        }
        System.out.println(player > computer ? "You win " + player + "" + computer
                : player < computer ? "Computer wins " + computer + "" + player
                        : "Overall tie " + player + "" + computer);
    }
}

```

*Code Screenshot 1*

TollBooth.java 1

Part A > Practical 1-A > TollBooth.java > ...

```java
import java.util.*;

public class TollBooth {
    record Vehicle(String number, String type) {
    }

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int total = 0, bike = 0, car = 0, truck = 0;
        while (true) {
            System.out.print("Vehicle number (done to finish): ");
            String number = sc.next().trim();
            if (number.equalsIgnoreCase("done"))
                break;
            System.out.print("Type (bike/car/truck): ");
            String type = sc.next().trim().toLowerCase();
            Vehicle v = new Vehicle(number, type);
            int toll = switch (v.type()) {
                case "bike" -> 20;
                case "car" -> 50;
                case "truck" -> 150;
                default -> 0;
            };
            if (toll == 0) {
                System.out.println("Unknown vehicle type");
                continue;
            }
            total += toll;
            switch (type) {
                case "bike" -> bike++;
                case "car" -> car++;
                case "truck" -> truck++;
            }
        }
        String frequent = bike >= car && bike >= truck ? "bike" : car >= truck ? "car" : "truck";
        System.out.println("Total toll: " + total);
        System.out.println("Most frequent: " + frequent);
    }
}

```

*Code Screenshot 2*

VendingMachine.java 1

Part A > Practical 1-A > VendingMachine.java > VendingMachine > main(String[])

```java
import java.util.Scanner;

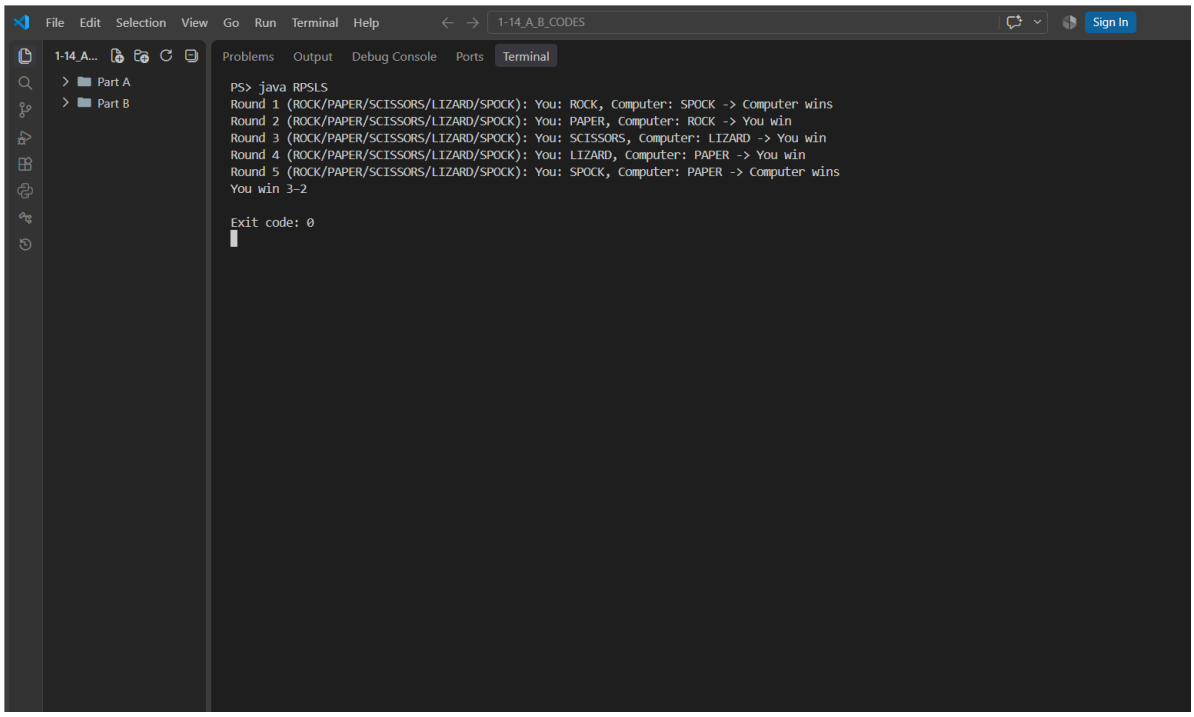
public class VendingMachine {
    enum Coin {
        ONE, TWO, FIVE, TEN
    }

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int total = 0, price = 15;
        while (total < price) {
            System.out.print("Enter coin (ONE, TWO, FIVE, TEN): ");
            Coin coin = Coin.valueOf(sc.next().trim().toUpperCase());
            total += switch (coin) {
                case ONE -> 1;
                case TWO -> 2;
                case FIVE -> 5;
                case TEN -> 10;
            };
            System.out.println("Total so far: " + total);
        }
        System.out.println("Paid. Change: " + (total - price));
    }
}

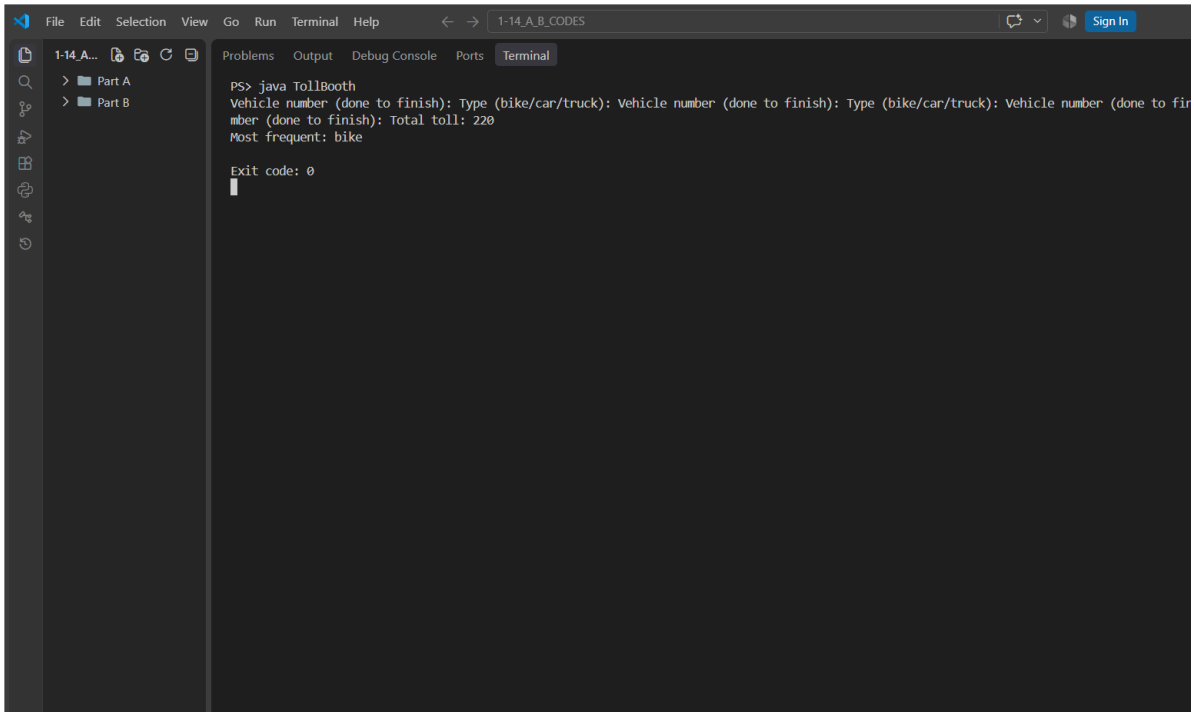
```

*Code Screenshot 3*

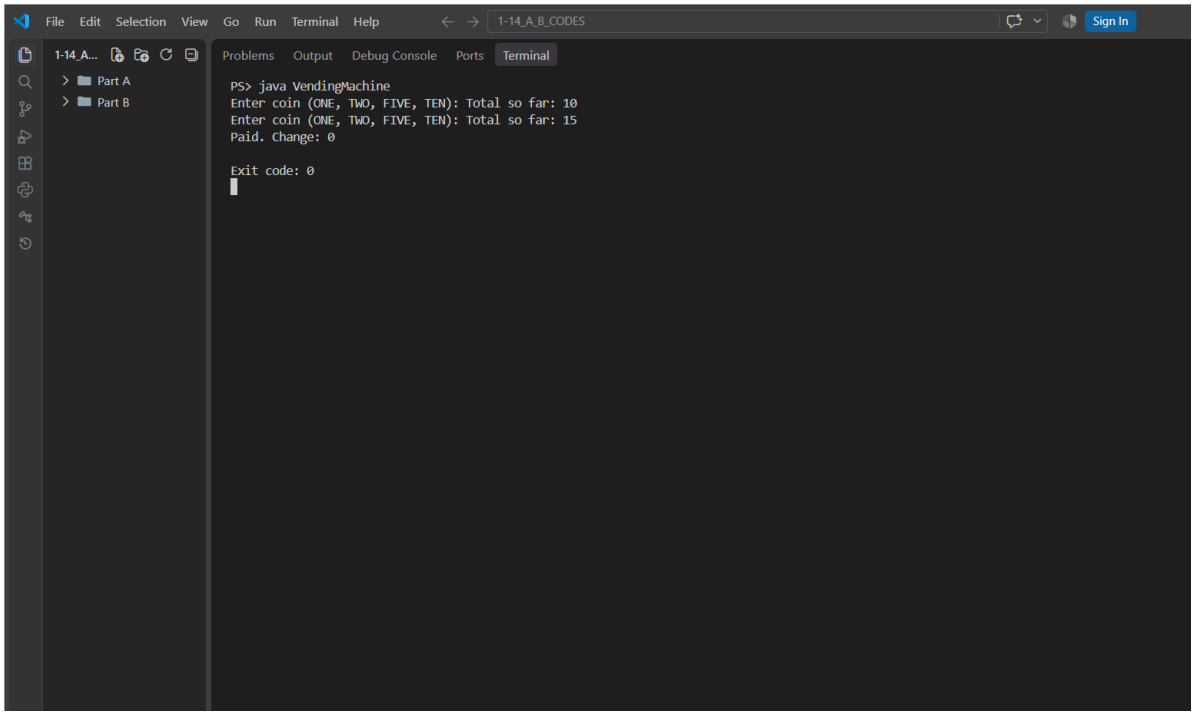
## OUTPUT SCREENSHOT:




*Output Screenshot 1*




*Output Screenshot 2*




*Output Screenshot 3*

## LEARNING OUTCOMES:

Develop a basic interactive Java console application and use enums, records, switch expressions, and input handling.